

Deep Learning (285652)

ClaimGuard: An Evidence-Backed Pipeline for Academic Citation Integrity Checking

Wagner Nils* Wenzel Nick†

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Submitted to Carsten Lanquillon

*212405, nwagner1@stud.hs-heilbronn.de

†207678, nwenzel@stud.hs-heilbronn.de

1. Abstract

Academic integrity checking goes beyond copied text: a genuine reference may fail to support its citing sentence. ClaimGuard identifies claim-bearing sentences and missing citations, validates references, retrieves evidence, and assigns five support labels. Its interface preserves the evidence trail and compares verifiers on identical retrieved input. It combines rules, scholarly metadata APIs, lexical or Sentence-BERT retrieval, local Ollama, an OpenAI frontier model, a SciFact LoRA classifier, and optional Fast-DetectGPT scoring. On a 30-case benchmark, lexical verification achieved 0.900 accuracy and Macro-F1; heuristic, Ollama, OpenAI, and LoRA achieved Macro-F1 values of 0.868, 0.728, 0.518, and 0.172. On 50 annotated report sentences, claim-type accuracy fell to 0.520 and citation-needed F1 to 0.267. Thirty references produced 0.467 final-status accuracy; DOI resolution was correct in seven cases, while retraction recall was 0.500. On 90 aligned SciFact pairs, LoRA beat majority (0.367 versus 0.167 Macro-F1) but trailed Ollama (0.433). Fast-DetectGPT reached 0.696 F1 and 0.924 AUROC on 15 human and 15 GPT-4 PubMed passages, but missed seven generated passages. ClaimGuard is reviewer triage, not an automatic misconduct detector. Limitations include single-annotator data, synthetic evidence, access bias, public-API limits, and model non-determinism.

Keywords: academic integrity, citation verification, retrieval-augmented generation, scientific claim verification, LoRA, human-in-the-loop review

2. Introduction

Academic reports increasingly combine conventional references, web-accessible research, and language-model-assisted writing. A bibliography may contain malformed or fabricated records, but even a genuine paper may not establish the claim attached to it. Plagiarism detection addresses textual overlap; it does not determine whether a sentence needs a citation, whether the cited work exists, or whether available evidence supports the sentence. Manually answering these questions requires repeated movement between a report, bibliographic databases, and source texts, which encourages superficial spot checks as documents grow.

The risk is asymmetric. Missing a misleading citation weakens assessment quality, while a false positive may unfairly imply misconduct. ClaimGuard consequently frames its output as decision support. It prioritizes passages for inspection and exposes the sentence, citation marker, parsed metadata, retrieved evidence, verifier, confidence, latency, and rationale. No score is converted into an accusation.

The project investigates four questions: (RQ1) Can lightweight section-aware rules identify claim types and missing citations in authentic PDFs? (RQ2) Can IEEE-like references be recovered and validated despite PDF extraction errors? (RQ3) How well do lexical and semantic retrieval support a five-way source-support judgment? (RQ4) What trade-offs emerge among a transparent heuristic, local Ollama, an OpenAI frontier model, and a SciFact-trained LoRA

adapter?

All three core modules were implemented: claim and citation analysis, reference parsing and validation, and retrieval-based support verification. Optional work comprises interchangeable verifier backends, Sentence-Transformer retrieval, a GPU-trained LoRA adapter, Fast-DetectGPT paragraph scoring, and Streamlit comparison. Fast-DetectGPT was evaluated on paired original and GPT-4 PubMed passages; Binoculars exceeded the 8 GB GPU. The contribution is an inspectable connection between citation need, bibliographic identity, evidence, and alternative verifiers.

3. Related Work

SciFact introduced expert-written scientific claims with support, refutation, and evidence rationales [1]. MultiVerS extended scientific verification with weak supervision and full-document context [2]. Both show why absence of evidence must remain distinct from contradiction. ClaimGuard adds upstream citation-need detection and bibliographic identity checks to this verification setting.

Citation-needed detection is a distinct upstream task. Redi et al. developed a citation-need taxonomy and models for Wikipedia verifiability [13]; their domain and editorial conventions differ from student reports. Sarol et al. paired retrieval with verification to study subtle citation-integrity errors in biomedical publications [14]. ClaimGuard targets the same claim–evidence relationship but adds local document ingestion and metadata validation.

Retrieval-augmented generation links models to external evidence [3], while Sentence-BERT supports efficient semantic retrieval [4]. GROBID and S2ORC provide document and bibliographic infrastructure [5, 6]; OpenAlex and Crossref expose complementary scholarly metadata [7, 8]. scite classifies citation contexts in indexed literature [9], whereas ClaimGuard starts from an arbitrary local report and connects its sentence to evidence.

LoRA enables parameter-efficient adaptation [10]. Binoculars uses cross-model perplexity, while Fast-DetectGPT uses conditional probability curvature [11, 12]. Paraphrasing and distribution shift undermine detector reliability [15]; broad tool tests likewise caution against treating such scores as misconduct evidence [16]. ClaimGuard therefore treats AI detection as optional risk signaling. Its gap is integration: citation need, bibliographic identity, retrieval, verifier disagreement, and a reviewer-facing evidence trail in one workflow.

4. System Architecture and Methods

4.1. Pipeline

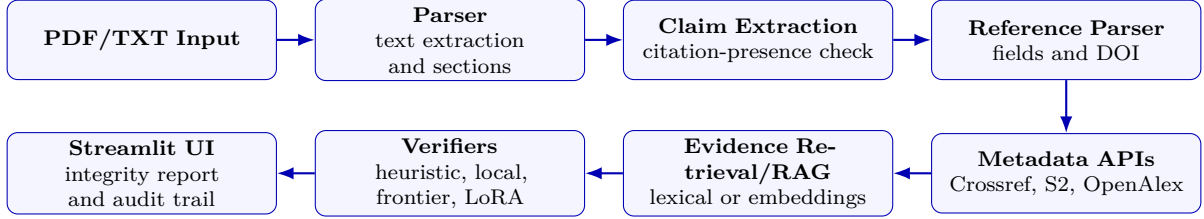


Figure 1: ClaimGuard architecture and data flow. Evidence, confidence, latency, and backend metadata remain attached to the final report.

Core module	Implemented components	Output	Evaluation evidence	Limitations
1: Claim extraction and citation presence	Section-aware sentence types, citation linking, inherited citations, severity rules	Sentence label, cited indices, missing-citation flag/severity	50 real sentences: accuracy .520, Macro-F1 .437; citation-needed F1 .267	Single annotator; rules confuse definitions, methods, and interpretations
2: Bibliographic validation	IEEE-like parser; identity/-DOI matching; Crossref, Semantic Scholar, OpenAlex; retraction flags	Parsed record, candidate, DOI, confidence, validation/problem status	30 references: parser 1.000; status Acc. .467/F1 .412; DOI 7/7; retraction F1 .667	Manual small gold set; 26/30 degraded API coverage; two retraction positives
3: Deep claim-source verification/RAG	Chunking, lexical/embedding retrieval, heuristic/Ollama/OpenAI/LoRA interfaces	Evidence chunks, five-way verdict, rationale, backend metadata	30 controlled cases: Macro-F1 heuristic .868, Ollama .728, OpenAI .518, LoRA .172	Synthetic short passages; incomplete full text; monetary cost not logged

Table 1: Core Module Compliance. Metrics refer only to the listed saved evaluations.

The parser accepts PDF and plain text, cleans extraction artifacts, separates body, references, and supplied evidence, and propagates section headings. Sentence splitting protects initials, abbreviations, decimal values, and IEEE citation markers. The claim module assigns one of five types: factual claim, methodological statement, opinion/interpretation, background/definition, or non-claim. Rules combine empirical verbs, numerical patterns, method cues, hedging, and definitional constructions. A factual sentence outside methods or results is flagged when it lacks a same-sentence or safely inherited citation; severity rises for numerical, causal, universal, or explicit prior-work assertions.

The reference module extracts index, authors, title, venue, year, and DOI and repairs common PDF artifacts. A parsed DOI is normalized and used for DOI-specific Crossref, Semantic Scholar, and OpenAlex lookups before title search; matching scores combine DOI, title, author, year, and venue. DOI resolution alone cannot prove identity: after a benchmark exposed chimeric citations, exact-DOI boosting was constrained by title and year consistency. Crossref retraction relations/subtypes and OpenAlex’s `is_retracted` field can produce `retracted_or_problematic`.

Coverage remains partial because Semantic Scholar supplies no retraction flag, DOI resolution is not a complete registry audit, and public services can throttle queries. Candidate metadata remains visible rather than being silently accepted.

Evidence from abstracts, supplied passages, metadata, or accessible full text is split into overlapping chunks. Lexical retrieval normalizes tokens, removes stop words, lightly stems terms, and scores overlap. Semantic retrieval encodes claims and chunks with `all-MiniLM-L6-v2` and normalized cosine similarity. Retrieval is restricted to the cited reference when possible, reducing contamination from unrelated works.

4.2. Verifier backends and interface

The deterministic verifier uses coverage, negation, directional conflict, and explicit absence-of-evidence cues. Ollama sends the same claim and top evidence to `qwen3:1.7b` at temperature zero. The OpenAI backend uses `gpt-5.4-mini`, disables response storage, and requires structured JSON. Both return exactly one of *supported*, *partially supported*, *not supported*, *contradicted*, or *insufficient evidence*, plus confidence and rationale. The shared prompt instructs models to judge only the supplied evidence; its full text and a sanitized response are in Appendix A.

The local SciFact classifier starts from `distilbert-base-uncased` with rank-8, alpha-16 LoRA adapters on query/value projections. Training used 1,261 pairs, 450 validation pairs, batch size 16, three epochs, and 30.8 seconds on an RTX 3070 Ti. The adapter contains 740,355 parameters (1.09% of 67,696,134 loaded parameters) and occupies 2.83 MiB. A seeded balanced subset (30 per native label) compares it directly with temperature-zero `qwen3:1.7b` prompting and a majority baseline on identical claim–abstract pairs. Its three native labels map to only three of ClaimGuard’s five categories.

4.3. Evaluation design

Evaluation is separated into authentic-document and controlled evidence. The authentic claim set contains ten manually selected sentences from each of five reports ($N = 50$), labeled once for claim type and citation need. Module 2 uses 20 authentic references plus ten controlled perturbations. Controlled Module 1 tests contain 25 patterns; Module 3 uses 30 balanced synthetic claim–evidence pairs. All four verifiers received identical retrieved chunks. LoRA validation uses all 450 SciFact pairs; its aligned comparison uses a reproducible balanced 90-pair subset. Module 4 uses 15 original PubMed answers and their 15 paired GPT-4 generations from the official Fast-DetectGPT repository (commit 971b052, seed 42; both members at least 40 words). This set provides provenance labels but no citation markers, so detector–citation correlation is not evaluated.

5. Results

5.1. Quantitative results

Component	<i>N</i>	Acc.	Prec.	Recall	F1
<i>Synthetic/controlled regression evidence</i>					
Synthetic citation-needed detection	25	1.000	1.000	1.000	1.000
Five-way verification, lexical	30	0.900	0.905	0.900	0.900 ^a
Five-way verification, embeddings	30	0.867	0.900	0.867	0.868 ^a
SciFact LoRA validation	450	0.409	–	–	0.353 ^a
SciFact majority baseline	90	0.333	–	–	0.167 ^a
SciFact LoRA, aligned subset	90	0.389	–	–	0.367 ^a
SciFact Ollama zero-shot	90	0.533	–	–	0.433 ^a
Module-2 final status	30	0.467	–	–	0.412 ^a
DOI resolution	7	1.000	1.000	1.000	1.000
Retraction flag	30	0.967	1.000	0.500	0.667
Fast-DetectGPT, PubMed/GPT-4	30	0.767	1.000	0.533	0.696
<i>Authentic-document evidence</i>					
Real claim-type classification	50	0.520	–	–	0.437 ^a
Real factual-claim detection	50	–	0.455	0.833	0.588
Real citation-needed detection	50	–	0.250	0.286	0.267
Reference single-record parsing	30	1.000	–	–	1.000 ^b

^aMacro-F1; ^bone parsed record per input. Module-2 gold labels were manually assigned.

Table 2: Saved evaluation results, explicitly separated by evidence type.

The dominant finding is the synthetic-to-real gap. Handcrafted patterns were perfect on the 25-case regression set, yet real-report claim accuracy was 0.520. Definitions were never recovered correctly as a class, and nine of fifteen methodological statements became factual claims. Citation-needed recall was 0.286: five of seven annotated omissions were missed. Rules remain useful for prioritization but do not generalize sufficiently for automatic enforcement.

Fast-DetectGPT achieved .924 AUROC with 1.019 s mean API latency. It produced no false positives on the 15 human passages, but detected only 8/15 GPT-4 passages. Thus its high precision came with limited recall on this small unedited set; it is a triage signal rather than authorship evidence. The benchmark contains no citation markers, so no association between detector score and citation behavior can be inferred.

Gold \ predicted	Factual	Method	Opinion	Definition	Non-claim
Factual claim	15	0	2	0	1
Methodological statement	9	6	0	0	0
Opinion/interpretation	4	0	5	0	1
Background/definition	5	0	0	0	2

Table 3: Confusion matrix for the 50 authentic sentences. No gold examples were labeled non-claim.

Module-2 task	N	Prec.	Recall	F1/Acc.	Notes
Single-record parsing	30	–	–	1.000 Acc.	20 authentic + 10 controlled references
Exists/not found	30	1.000	.538	.700 F1	Conservative: 12 false negatives, no false positives
Final four-way status	30	–	–	.412 Macro-F1	Accuracy .467; partial-match F1 .000
DOI resolution	7	1.000	1.000	1.000 F1	Five valid, two invalid; chimeric identity checked separately
Retraction flag	30	1.000	.500	.667 F1	Only two positive cases; one was missed

Table 4: Gold-labeled Module-2 evaluation. API coverage was degraded for 26/30 cases, so status metrics are service-condition dependent.

On the balanced support benchmark, lexical verification reached 0.900 Macro-F1 and embeddings 0.868. On the aligned SciFact subset, LoRA improved over majority by .200 Macro-F1 (.367 versus .167) but trailed Ollama zero-shot (.433). LoRA nevertheless recovered CONTRADICTION with .360 F1, whereas Ollama never correctly predicted that class (F1 .000), instead mapping 21/30 contradictions to SUPPORT. LoRA’s five-label transfer remained weak (.172 Macro-F1) because it cannot emit *partially supported* or *not supported*. Fine-tuning therefore added class-specific signal, but did not produce an overall zero-shot improvement and did not transfer cleanly across taxonomies.

5.2. Qualitative behavior

Case	Observed behavior	Interpretation
RQ5 sentence 12	Flagged an uncited claim that retrieval is a central RAG bottleneck.	Useful high-severity triage.
Report-5 references	Recovered five titles, years, and venues despite collapsed spacing.	Parser repair succeeded on authentic PDF text.
Contradiction C1	Matched “do not eliminate errors” against “eliminate all hallucinations.”	Negation and direction cues worked.
RQ1-02	Labeled a result interpretation as a factual claim.	Mixed rhetorical roles confuse rules.
RQ5 title page	Flagged header text as an uncited factual claim.	Layout noise survived extraction.

Table 5: Five representative successes and failures.

5.3. Local versus frontier comparison

Backend	N	Acc.	Macro-F1	Mean latency	Tokens / cost	Typical failure
Heuristic	30	.867	.868	0 ms	no tokens	partial → supported (2)
Ollama qwen3:1.7b	30	.733	.728	2744.5 ms	3775 prompt / 1674 generated; cost not logged	partial → supported (3)
OpenAI gpt-5.4-mini	30	.633	.518	1424.1 ms	6050 input / 1466 output; cost not logged	not supported → insufficient (5)
SciFact LoRA	30	.300	.172	6.0 ms	no tokens	partial → insufficient (6)

Table 6: Four-backend comparison on the same 30 gold-labeled cases and identical retrieved evidence. Model calls occurred on 27 cases; three no-evidence cases abstained before invocation.

Ollama/OpenAI agreement was .567 (17/30). The local model exceeded the frontier model by .210 Macro-F1, while OpenAI was faster on this hardware/network run. OpenAI’s main error was collapsing *not supported* into *insufficient evidence*; Ollama over-predicted full support for partial evidence. This controlled benchmark provides no case where frontier quality justifies its external-service dependency or unlogged monetary cost. That conclusion is benchmark-specific, not a general model ranking.

5.4. Metadata-search and internal baselines

#	Queried Report-5 work (short title)	Score	Crossref top-1 judgment
1	Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks	.593	Incorrect later QA variant
2	Sentence-BERT	.867	Correct
3	RAG for Large Language Models: A Survey	.780	Incorrect graph-RAG variant
4	Lost in the Middle	.779	Incorrect “Found in the Middle”
5	LLMs Distracted by Irrelevant Context	.459	Incorrect unrelated security work

Table 7: Five-case Crossref metadata-search baseline. Scores are fuzzy title similarity.

Crossref’s top candidate was exact for one of five records. Two incorrect near-neighbors exceeded the nominal 0.75 fuzzy threshold because title terms and year overlapped. This metadata-search baseline does not evaluate citation need or claim–source support. A checkpoint-capable OpenAI-without-RAG ablation was implemented, but its 30-case run was not completed after a provider timeout and usage-limit rejection; no metric is reported. No completed scite, SemanticCite, or other external-tool prediction file exists; the CSV is only a template. A full existing-tool

benchmark therefore remains future work. ClaimGuard is an orchestration layer over scholarly services, not their replacement.

5.5. Answers to the research questions

RQ	Evidence-grounded answer
RQ1	Only partially. On 50 authentic sentences, claim-type accuracy was .520 (Macro-F1 .437) and citation-needed F1 was .267; lightweight rules are triage, not comprehensive detection.
RQ2	Partially. The 30-reference benchmark parsed one record per input and resolved all seven DOI checks, but final-status accuracy was .467 under degraded API coverage; retraction recall was .500 on two positives.
RQ3	On 30 controlled cases, lexical verification reached .900 Macro-F1 and embeddings .868. Because evidence was short and synthetic, this does not establish real-document verification accuracy.
RQ4	On five-label inputs, Macro-F1 was .868 heuristic, .728 Ollama, .518 OpenAI, and .172 transferred LoRA. On 90 native-label SciFact pairs, LoRA (.367) beat majority (.167) but not Ollama zero-shot (.433).

Table 8: Direct answers to RQ1–RQ4.

6. Discussion

ClaimGuard’s strongest property is its end-to-end audit trail: a reviewer can move from sentence and citation marker to metadata candidate, retrieved passage, backend, and rationale. The main empirical lesson is nevertheless caution. Strong controlled verification (.900 Macro-F1) coexists with weak real-document claim transfer (.437 Macro-F1), so synthetic regression scores must not be read as deployment performance.

Failure mode	Cause	Example/evidence	Mitigation
Layout false positive	PDF headers and title text enter sentence stream	RQ5 title/header flagged as an uncited claim	Layout-aware parsing; exclude front matter
Claim-type confusion	Mixed rhetorical roles and narrow rules	9/15 method statements predicted factual	Learned section-aware model with abstention
Metadata/API failure	Near-neighbors and throttled services	Status Acc. .467; degraded coverage in 26/30 cases	Require identity agreement; cache/backoff; show candidates
Access bias	Paywall/non-OA or abstract-only evidence	No source-level rate measured; missing full text can yield insufficient evidence	Separate access failure from support; licensed/OA retrieval
Taxonomy/model bias	SciFact has three labels; prompts may avoid contradiction	LoRA transfer F1 .172; Ollama contradiction F1 .000	Align labels; class-balanced prompts/data

Table 9: Observed or explicitly unmeasured failure modes and mitigations.

Access bias deserves special emphasis: paywalled or non-open-access works are harder to verify than sources with complete machine-readable text. Consequently, *insufficient evidence* may describe retrieval or access limits rather than an unsupported claim, and may systematically favor open-access literature. API timeouts and incomplete deposits add similar missingness. The evaluation is small and single-annotator; Module 2 has only seven DOI checks and two retraction positives, while the local/frontier comparison uses synthetic short evidence.

False positives are consequential in academic assessment. Common knowledge, inherited citations, extraction errors, and cautious scientific language can all trigger incorrect flags. AI-text scores are particularly unsuitable as authorship proof [15, 16]. ClaimGuard therefore avoids misconduct labels and requires source inspection. Fast-DetectGPT’s perfect precision on 15 human passages must not be generalized: it missed 7/15 generated passages, and edited or paraphrased text was not tested. Citation association also remains unmeasured because the provenance-labeled benchmark contains no citation markers; private student reports were not transmitted to the external API for an exploratory analysis. Future work should add double annotation, authentic claim–source labels, broader DOI/retraction cases, an external-tool benchmark, and a consented, preregistered AI-detection test including edited/paraphrased text and citation behavior.

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A. Reproducibility and Prompt Details

A.1. Environment and saved outputs

Experiments used Python 3.14.3, PyTorch 2.11.0 with CUDA 12.8, an NVIDIA RTX 3070 Ti with 8 GB VRAM, and seed 42 for LoRA training. Primary artifacts are:

- `outputs/evaluation/rq_claim_evaluation.json`
- `outputs/evaluation/reference_evaluation.json`
- `outputs/evaluation/evaluation_lexical.json`
- `outputs/evaluation/evaluation_embeddings.json`
- `outputs/evaluation/evaluation_lora.json`
- `outputs/evaluation/model_comparison_full.json`
- `outputs/evaluation/module2_validation.json`
- `outputs/evaluation/lora_artifact_summary.json`
- `outputs/evaluation/scifact_lora_vs_zeroshot.json`
- `outputs/evaluation/fast_detect_gpt.json`
- `outputs/evaluation/README.md`
- `outputs/evaluation/crossref_baseline_rq5.json`
- `outputs/models/scifact-lora/training_report.json`

API identifiers are retained locally; credentials are excluded.

A.2. Verifier prompt

Classify the claim using exactly one status: supported, partially_supported, not_supported, contradicted, or insufficient_evidence. Judge only from the supplied evidence. Return JSON with status, confidence (0..1), and a concise rationale.

A.3. LoRA and execution details

The adapter used rank 8, alpha 16, query/value projections, an effective batch size of 16, three epochs, and 740,355 saved trainable parameters (1.09% of 67,696,134 loaded parameters). Training took 30.8 seconds; `adapter_model.safetensors` is 2,965,276 bytes (2.83 MiB). Reproduction commands and response samples are stored with the project outputs.